Supplementary Material:
POLARIS: Polarity- and Selectivity-Controlled Graph Attention for Heterophilous Graph Neural Networks

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The appendices are organised as follows. **Appendix A** gives the full proofs of every theoretical result in the main paper. **Appendix B** lists the complete hyperparameter, protocol, and reproducibility settings. **Appendix C** reports additional experiments and analyses (extra baselines, adaptive variants, calibration detail, and a study of the label-safe bias), all under the same protocol. Theorem-like results restate the main-paper statements; the prefix "A" marks the supplement's own numbering, and each restatement names the corresponding main-paper result. As in the main paper, **bold** and underline mark the best and second-best of the compared entries in every table.

A Proofs

A.1 Setup and the averaging baseline

We recall the signal model used throughout the analysis.

Definition A1 (Symmetric K-class model (main-paper Sec. 4))**.** The K class means $\{\mu_k\}_{k=1}^K \subset \mathbb{R}^d$ form a simplex equiangular tight frame: $\|\mu_k\|^2 = s$, $\langle \mu_k, \mu_l\rangle = -s/(K-1)$ for $k \neq l$, and $\sum_k \mu_k = 0$. Node j has feature $x_j = \mu_{c_j} + \varepsilon_j$ with zero-mean noise independent across nodes. For a target node i of class c, each neighbour $j \in \mathcal{N}(i)$ is same-class with probability h (the *edge homophily*) and otherwise belongs to one of the other $K-1$ classes uniformly (probability $(1-h)/(K-1)$ each).

Lemma A1 (Expected neighbour signal (main-paper Sec. 4))**.** *For $j \in \mathcal{N}(i)$, $\mathbb{E}[x_j \mid c] = \gamma_K(h)\,\mu_c$ with* $\gamma_K(h) = \dfrac{Kh-1}{K-1}$.

Proof. $\mathbb{E}[x_j \mid c] = h\,\mu_c + \frac{1-h}{K-1}\sum_{l\neq c}\mu_l$. Since $\sum_l \mu_l = 0$ we have $\sum_{l\neq c}\mu_l = -\mu_c$, so $\mathbb{E}[x_j \mid c] = \big(h - \frac{1-h}{K-1}\big)\mu_c = \frac{h(K-1)-(1-h)}{K-1}\mu_c = \frac{Kh-1}{K-1}\mu_c$. □

Proposition A1 (Averaging degrades, then misleads; main-paper Prop. 1)**.** *Under uniform attention $m_i = \frac{1}{|\mathcal{N}(i)|}\sum_j x_j$, $\mathbb{E}[m_i \mid c] = \gamma_K(h)\,\mu_c$: correctly aligned with class c iff $h > \frac{1}{K}$, with* zero *expected signal at $h = \frac{1}{K}$, and for $h < \frac{1}{K}$ negatively aligned with μ_c and positively with every other class mean ($\langle \mathbb{E}[m_i \mid c], \mu_c\rangle < 0$ while $\langle \mathbb{E}[m_i \mid c], \mu_l\rangle > 0$ for $l \neq c$), so a class-mean readout ranks the true class* last.

Proof. By linearity and Lemma A1, $\mathbb{E}[m_i \mid c] = \gamma_K(h)\mu_c$; the sign of $\gamma_K(h) = (Kh-1)/(K-1)$ is that of $Kh-1$. The class-mean scores $\langle \gamma_K\mu_c, \mu_k\rangle$ equal $\gamma_K s$ for $k = c$ and $-\gamma_K s/(K-1)$ for $k \neq c$; when $\gamma_K < 0$ ($h < 1/K$) the former is negative and the latter positive. □

A.2 How POLARIS repairs the signal

Proposition A2 (Convex self-mixing gives a correct-signal floor; main-paper Prop. 2)**.** *With convex self–neighbour mixing $u_i = \lambda x_i + (1-\lambda)m_i$ and uniform neighbour attention, $\mathbb{E}[u_i \mid c] = \big[\lambda + (1-\lambda)\gamma_K(h)\big]\mu_c$, correctly signed iff $\lambda > \frac{1-Kh}{K(1-h)}$. In particular $\lambda \geq \frac{1}{K}$ guarantees correct-class alignment for* every *$h \in [0,1)$, the threshold being exactly $\frac{1}{K}$ at full heterophily $h = 0$.*

Proof. Since $\mathbb{E}[x_i \mid c] = \mu_c$, Prop. A1 gives the coefficient $\lambda + (1-\lambda)\gamma_K(h)$, positive iff $\lambda(1-\gamma_K) > -\gamma_K$. With $1-\gamma_K = K(1-h)/(K-1) > 0$ and $-\gamma_K = (1-Kh)/(K-1)$, this is $\lambda > \frac{1-Kh}{K(1-h)}$ for $h < 1/K$ (vacuous for $h \geq 1/K$, where $\gamma_K \geq 0$). The right-hand side falls from $1/K$ at $h=0$ to 0 at $h = 1/K$, so $\lambda \geq 1/K$ suffices throughout. □

Proposition A3 (Selectivity recovers magnitude; main-paper Prop. 3)**.** *If attention places total mass p on same-class neighbours and spreads the rest uniformly over the others, $\mathbb{E}[m_i \mid c] = \gamma_K(p)\,\mu_c$, which strictly exceeds the uniform case ($p = h$ in expectation) whenever $p > h$, is correctly signed for $p > \frac{1}{K}$, and is maximised as the attention entropy $H_i \to 0$ ($p \to 1$).*

Proof. Grouping the attention weight by class and applying Lemma A1 with h replaced by the realised same-class mass p gives $\mathbb{E}[m_i \mid c] = \big(p - \frac{1-p}{K-1}\big)\mu_c = \gamma_K(p)\mu_c$, strictly increasing in p. Uniform attention has $p = h$ in expectation, recovering Prop. A1; concentrating mass on same-class neighbours raises p toward 1 as $H_i \to 0$. □

Remark 1 (Two-class special case). At $K = 2$ the means are antipodal ($\mu_0 = -\mu_1$), $\gamma_2(h) = 2h-1$, threshold $\frac{1}{2}$, and $\lambda \geq \frac{1}{2}$: averaging exactly inverts the class signal below $h = \frac{1}{2}$, the canonical heterophily picture. The general threshold $\frac{1}{K}$ is the random-neighbour level.

Proposition A4 (Selectivity trades variance for bias)**.** *Let neighbour features be $h_j = \mu_{c_j} + \varepsilon_j$ with iid noise $\mathbb{E}[\varepsilon_j] = 0$, $\mathrm{Cov}(\varepsilon_j) = \sigma^2 I$. Then $m_i = \sum_j \alpha_{ij} h_j$ has bias $\mathbb{E}[m_i \mid c] = \gamma_K(p)\,\mu_c$ (Prop. A3) and variance*

$$\mathrm{Var}(m_i) = \sigma^2 \textstyle\sum_j \alpha_{ij}^2.$$

The collision mass $\sum_j \alpha_{ij}^2 \in [1/d_i, 1]$ (d_i the neighbourhood size) is minimised by uniform attention (maximal entropy, value $1/d_i$) and increases monotonically as attention concentrates ($\to 1$ as $H_i \to 0$). Hence lower aggregation entropy raises *the aggregate variance.*

Proof. Independence gives $\mathrm{Var}(m_i) = \sum_j \alpha_{ij}^2 \mathrm{Var}(\varepsilon_j) = \sigma^2 \sum_j \alpha_{ij}^2$. On the simplex $\{\alpha \geq 0,\ \sum_j \alpha_{ij} = 1\}$ over d_i neighbours, $\sum_j \alpha_{ij}^2$ is convex, minimised at $\alpha_{ij} = 1/d_i$ (value $1/d_i$) and maximised at a vertex (value 1); it equals $e^{-H_2(\alpha_i)}$ for the Rényi-2 entropy H_2, hence is decreasing in the Shannon entropy H_i along the temperature path. □

Remark 2 (Accuracy-optimal entropy tracks homophily; the regulariser sets the operating point). Combining the bias (Prop. A3) and the variance (Prop. A4): under homophily the bias is already correct ($\gamma_K(h) > 0$), so accuracy is variance-limited and best at *high* entropy (broad averaging); under heterophily the bias is wrong, so *low* entropy (or a sign) is needed despite the variance cost. The accuracy-optimal entropy thus increases with homophily, matching the measured trend (main paper). The entropy regulariser sets the entropy away from this free optimum: on homophilous graphs that lowers accuracy but can improve calibration (App. C.16), as less averaging retains intra-class representation diversity and softens over-confident logits. POLARIS-E is therefore a controllable accuracy–calibration operating point, not a free gain.

Proposition A5 (General class means: the threshold without ETF)**.** *The ETF assumption yields the clean value $1/K$ but is not required for the phenomenon. For* arbitrary *class means $\{\mu_k\}$, the same neighbour model gives $\mathbb{E}[m_i \mid c] = h\,\mu_c + (1-h)\,\bar{\mu}_{\neg c}$ with $\bar{\mu}_{\neg c} = \frac{1}{K-1}\sum_{l \neq c} \mu_l$ the off-class centroid. Let $\rho_c = \langle \bar{\mu}_{\neg c}, \mu_c \rangle / \|\mu_c\|^2$ measure its alignment with μ_c. Then the component of the aggregate along μ_c has coefficient $h + (1-h)\rho_c$, so averaging is correctly aligned with class c iff $h > h_c^\star := \frac{-\rho_c}{1-\rho_c}$ (a non-trivial threshold whenever $\rho_c < 0$, i.e. the competing centroid opposes μ_c; if $\rho_c \geq 0$ averaging never misleads class c).*

Proof. Linearity gives $\mathbb{E}[m_i \mid c] = h\mu_c + (1-h)\bar{\mu}_{\neg c}$. Projecting onto μ_c, the coefficient is $\langle \mathbb{E}[m_i \mid c], \mu_c \rangle / \|\mu_c\|^2 = h + (1-h)\rho_c$, positive iff $h(1-\rho_c) > -\rho_c$, i.e. $h > -\rho_c/(1-\rho_c)$ (using $1 - \rho_c > 0$). Under the ETF model $\bar{\mu}_{\neg c} = -\mu_c/(K-1)$, so $\rho_c = -1/(K-1)$ and $h_c^\star = 1/K$ for every class, recovering Prop. A1. □

Thus the misleading-averaging threshold is a generic property of separated class means; the ETF model only collapses the class-wise thresholds $h_c^\star$ to the single value $1/K$. Self-mixing and selectivity carry over unchanged with $\gamma_K(h)$ replaced by $h + (1-h)\rho_c$.

Proposition A6 (Temperature is a monotone, surjective entropy control; main-paper Prop. 4)**.** *Fix logits $z \in \mathbb{R}^k$ that are not all equal and let $\alpha(\tau) = \mathrm{softmax}(z/\tau)$ with entropy $H(\tau) = -\sum_j \alpha_j \log \alpha_j$. Then H is strictly increasing on $(0, \infty)$ with $\lim_{\tau \to 0^+} H = 0$ and $\lim_{\tau \to \infty} H = \log k$. Consequently, with a bounded temperature $\tau \in [\tau_{\min}, \tau_{\max}]$, the attainable entropies form the interval $[H(\tau_{\min}), H(\tau_{\max})]$ and every target therein is realised.*

Proof. Let $\beta = 1/\tau > 0$ and $p_j = e^{\beta z_j}/Z$ with $Z = \sum_j e^{\beta z_j}$. The Gibbs entropy can be written $H = \log Z - \beta\,\mathbb{E}_p[z]$, where $\mathbb{E}_p[z] = \sum_j p_j z_j$. Differentiating in β and using the standard exponential-family identities $\frac{d}{d\beta} \log Z = \mathbb{E}_p[z]$ and $\frac{d}{d\beta}\mathbb{E}_p[z] = \mathrm{Var}_p(z)$,

$$\frac{dH}{d\beta} = \mathbb{E}_p[z] - \mathbb{E}_p[z] - \beta\,\mathrm{Var}_p(z) = -\beta\,\mathrm{Var}_p(z) \leq 0,$$

with strict inequality whenever z is non-constant (then $\mathrm{Var}_p(z) > 0$). Hence H is strictly decreasing in β, i.e. strictly increasing in τ. As $\beta \to \infty$ ($\tau \to 0^+$) the mass concentrates on $\arg\max_j z_j$ so $H \to 0$; as $\beta \to 0$ ($\tau \to \infty$) the distribution becomes uniform so $H \to \log k$. Strict monotonicity and continuity of H on $[\tau_{\min}, \tau_{\max}]$ then give, by the intermediate value theorem, that the image is exactly $[H(\tau_{\min}), H(\tau_{\max})]$ and every value in it is attained. □

A.3 Depth stability

Theorem A1 (Non-expansive depth stability, unsigned and signed; main-paper Thm. 1)**.** *Let a POLARIS layer act as $H^{(l+1)} = T^{(l)} H^{(l)}$ with $T^{(l)} = \lambda I + (1-\lambda) A^{(l)}$ and $\lambda \in [0,1]$, where $A^{(l)}$ has absolutely sub-stochastic rows, $\sum_j |A_{ij}^{(l)}| \leq 1$ for all i ($\star$). Then $\|T^{(l)}\|_\infty \leq 1$, and for any two inputs $H_1^{(0)}, H_2^{(0)}$ and any depth L,*

$$\big\|H_1^{(L)} - H_2^{(L)}\big\|_\infty \leq \big\|H_1^{(0)} - H_2^{(0)}\big\|_\infty.$$

Proof. By the triangle inequality each absolute row sum of $T^{(l)}$ obeys $\sum_j |T_{ij}^{(l)}| \leq \lambda \sum_j |I_{ij}| + (1-\lambda) \sum_j |A_{ij}^{(l)}| \leq \lambda + (1-\lambda) = 1$, using ($\star$) for the second term. The induced ∞-norm of a matrix is its maximum absolute row sum, so $\|T^{(l)}\|_\infty \leq 1$. Linearity gives $H_1^{(l+1)} - H_2^{(l+1)} = T^{(l)}(H_1^{(l)} - H_2^{(l)})$, so by submultiplicativity $\|H_1^{(l+1)} - H_2^{(l+1)}\|_\infty \leq \|H_1^{(l)} - H_2^{(l)}\|_\infty$; iterating over L layers yields the bound. Finally both variants satisfy ($\star$): the unsigned operator has $A_{ij} = \alpha_{ij} \geq 0$ with $\sum_j \alpha_{ij} = 1$, and the signed operator has $A_{ij} = \alpha_{ij}\psi_{ij}$ with $|\psi_{ij}| \leq 1$, whence $\sum_j |\alpha_{ij}\psi_{ij}| \leq \sum_j \alpha_{ij} = 1$. □

Lemma A2 (The gate keeps representations bounded; main–paper Lemma 1)**.** *For* any *coordinate-wise gate $g_i \in [0,1]^d$, possibly input-dependent, the gated update $h_i' = g_i \odot u_i + (1-g_i) \odot h_i$ satisfies $\|h_i'\|_\infty \leq \max(\|h_i\|_\infty, \|u_i\|_\infty)$.*

Proof. For each coordinate r, $h_{i,r}' = g_{i,r} u_{i,r} + (1-g_{i,r}) h_{i,r}$ with $g_{i,r} \in [0,1]$ is a convex combination of $u_{i,r}$ and $h_{i,r}$, hence lies between them: $|h_{i,r}'| \leq \max(|u_{i,r}|, |h_{i,r}|)$. Taking the maximum over r (and over nodes i) gives the claim.

The bound uses only $g_i \in [0,1]^d$ and so is unaffected by the gate being a learned function of the inputs. $\square$

Remark 3 (On metric non-expansiveness of the gated layer). Lemma A2 bounds the representation *norm*, which is what the depth guarantee needs: combined with $\|T^{(l)}\|_\infty \le 1$ for the realised neighbour operator (Thm. 1), it gives $\|H^{(L)}\|_\infty \le \|H^{(0)}\|_\infty$ at every depth, so representations cannot blow up regardless of how the attention and gate depend on the input. The two-input contraction $\|H'^{(1)} - H'^{(2)}\|_\infty \le \|H^{(1)} - H^{(2)}\|_\infty$ does *not* follow from convexity alone once $g_i^{(1)} \neq g_i^{(2)}$: the difference carries an extra term $(g_i^{(1)} - g_i^{(2)}) \odot (u_i - h_i)$. It is controlled because the gate $g_i = \sigma(W_g[h_i \| u_i])$ and the softmax attention are Lipschitz (compositions of bounded linear maps with 1-Lipschitz σ and softmax), giving the layer a finite Lipschitz constant; we do not require this constant to be ≤ 1 for the depth conclusion, which rests on the norm bound above.

A.4 Label safety, equivariance, and multi-class

Proposition A7 (Label safety and permutation equivariance; main-paper Prop. 5)**.** *(i) The discriminative term $\lambda_{\text{disc}}\nu_{ij}\delta_{ij}\mathbb{1}[train]$ is identically zero at inference, so the inference map $f(X,G)$ is independent of all labels. (ii) For any permutation matrix P, $f(PX, PGP^\top) = P\,f(X,G)$.*

Proof. (i) At inference the mode indicator $\mathbb{1}[\text{train}] = 0$, so the added logit term vanishes for every pair (i,j) regardless of ν_{ij}, δ_{ij}. Every remaining quantity (the projection $h = W^\top x$, the pairwise logits z_{ij}, the softmax weights α_{ij}, the mix u_i, and the gate g_i) is a function of (X,G) only. Hence $f(X,G)$ does not depend on any label, and in particular cannot leak training labels. (ii) Relabel nodes by a permutation matrix P. The shared W acts identically on every node, so projections permute as PH. Each logit z_{ij} depends only on the unordered pair of feature vectors (h_i, h_j) and on whether $(j,i) \in E$, both preserved under $PGP^\top$; the neighborhood softmax is computed over the permuted neighborhood $\tilde{\mathcal{N}}(\pi(i)) = \{\pi(j) : j \in \tilde{\mathcal{N}}(i)\}$, so $A \mapsto PAP^\top$. Then $T = \lambda I + (1-\lambda)A \mapsto PTP^\top$, and the propagated features transform as $PTP^\top\,(PH) = PTH$; the elementwise gate commutes with the row permutation. Composing layers and the (node-wise) classifier preserves equivariance, giving $f(PX, PGP^\top) = Pf(X,G)$. $\square$

B Hyperparameters, Protocol, and Reproducibility

All numbers in the main paper are produced by the released code with the settings below; no per-dataset tuning is applied unless stated.

B.1 Protocol and optimization

Models are trained with **Adam**, and checkpoints are selected on validation accuracy (strict improvement; ties keep the earlier epoch). Splits are used strictly as: *train* for the loss/gradients, *validation* for checkpoint selection, *test* for the reported metrics only. Table 1 lists the shared optimization settings.

Table 1: Shared optimization settings (all models).

Setting	Value
Optimizer	Adam
Learning rate	0.01
Weight decay	5×10^{-4}
Label smoothing	0.1
Dropout	0.5
Epochs (graphs)	120
Checkpoint	best val. acc. (strict $>$)
Splits	10 Geom-GCN (Cora: 1)
Random seed	split index $+ 1000r$ (repeat r)

Datasets. We use the heterophilous WebKB graphs (Texas, Wisconsin, Cornell) and Actor with the ten standard Geom-GCN train/val/test splits, and homophilous Cora with the standard public split as a regime contrast. Edge homophily h (fraction of edges joining same-label nodes) is reported alongside each dataset in the main paper. The large leakage-free benchmarks differ in task type, which matters when reading their metrics: *Roman-empire* is an 18-class problem and *Amazon-ratings* a 5-class problem (both reported by accuracy), whereas *Minesweeper* is a *binary*, *class-imbalanced* node-classification task ($\approx 20\%$ positives on a synthetic grid). On Minesweeper accuracy is therefore a weak metric (the majority-class baseline already scores high), so ROC–AUC is the headline number and the two should not be compared across datasets. AUC also varies sharply between methods there precisely because the task is structural: a model that fails to exploit the grid neighbourhood produces near-constant scores and lands at AUC ≈ 0.5 (chance), while one that does separates the classes (POLARIS reaches AUC 0.86). The accuracy/AUC gap is thus expected and reflects the metric, not an inconsistency.

B.2 POLARIS architecture and component settings

Table 2 lists the operator settings; the paragraphs below document the optional components (entropy regulariser, temperature, signed coefficient) and the rationale for their fixed choices.

Table 2: POLARIS operator settings. The bounded temperature is $\tau = \tau_{\min} + (\tau_{\max} - \tau_{\min})\sigma(\theta_\tau)$.

Setting	Value
Hidden width d	64
Depth L (layers)	4
Aggregator	sum
Attention nonlin.	LeakyReLU (0.2)
Temperature $[\tau_{\min}, \tau_{\max}]$	[0.25, 4.0]
λ, λ_{disc}	learned, $\sigma(\theta)$
Entropy reg. β (POLARIS/-U)	0

Entropy regulariser (POLARIS-E) settings and sensitivity. The entropy regulariser adds $\beta\,\frac{1}{L}\sum_l \frac{1}{n}\sum_i (H_i^{(l)} - H_i^\star)^2$ with degree-aware target $H_i^\star = \rho \log|\tilde{\mathcal{N}}(i)|$. We do *not* tune ρ per dataset: it is a single global parameter learned through $\rho = \sigma(\theta_\rho)$ (one scalar shared across layers), initialised at $\theta_\rho = 0$ ($\rho = 0.5$). The weight is fixed at $\beta = 1.0$ for every POLARIS-E run; we chose it once on Wisconsin validation and reused it everywhere. POLARIS-E is an *analysis* variant that makes the realised entropy a set quantity, not the accuracy configuration, so we keep its hyperparameters fixed rather than search them, and report (main text) that it is near-neutral for accuracy across the range we observed ($\beta \in [0.5, 2]$, ρ free).

Temperature: parametrisation and bounds. The temperature is *already per layer*: each POLARIS layer carries its own θ_τ, so a depth-L model learns L temperatures rather than a single global one. The bounds $[\tau_{\min}, \tau_{\max}] = [0.25, 4.0]$ were fixed once and not searched. We expect accuracy and calibration to be largely insensitive to them for a structural reason: the realised aggregation entropy is set jointly by τ and the spread of the attention logits, and the mechanism analysis (main text, Fig. 2) shows the operating point is reached mainly through the logit scale, with the learned $\tau \approx 2.0$ sitting in the interior of the interval across all datasets. Widening or narrowing $[\tau_{\min}, \tau_{\max}]$ therefore changes the *reachable* entropy range (Prop. 4) but not the operating point the model selects, so it should not materially move performance; the bound only matters if it excludes the self-selected level, which it does not here. A per-*node* temperature is supported by the same parametrisation (replace the scalar θ_τ by a node-wise head) but we did not adopt it: on these small graphs it adds capacity that risks overfitting, and the per-layer temperature already suffices. A systematic bounds/per-node sweep is left to future work.

Alternative signed mechanisms (not evaluated). We use the bounded tanh coefficient $\psi_{ij} = \tanh(c^\top [h_i \| h_j])$ and do *not* empirically compare alternative signed-normalisation schemes (sparsemax, α-entmax) or explicit high-pass spectral filters. As noted in the main related work, sparse-attention variants drive some weights to exactly zero but stay non-negative, so they sharpen selectivity without enabling repulsion; combining sparse normalisation with a sign is a natural design we leave to future work. The tanh coefficient is convenient here because $|\psi| \leq 1$ preserves the non-expansiveness guarantee (Thm. 1) directly, whereas a generic signed normalisation would have to re-establish it.

B.3 Calibration protocol

ECE uses 15 equal-width confidence bins on the test split, $\mathrm{ECE} = \sum_b \frac{|B_b|}{N} |\mathrm{acc}(B_b) - \mathrm{conf}(B_b)|$ with conf the mean top-class softmax probability. NLL is the mean test cross-entropy and the Brier score the mean squared error between the softmax vector and the one-hot label; all three are computed on the val-selected checkpoint, averaged over the same splits as accuracy, with no post-hoc temperature scaling so that the numbers reflect the operator's native calibration. We report native calibration deliberately: post-hoc temperature scaling fits one scalar per model on validation and is orthogonal to the operator, so it could be applied to every method and would not, on its own, establish which operator is better calibrated. The relevant comparison is that POLARIS is better calibrated *before* any such correction. Whether its advantage persists *after* temperature-scaling every baseline is a fair follow-up we did not run; because temperature scaling is monotone in the logits it leaves accuracy unchanged and typically shrinks all methods' ECE toward a common floor, so we would expect the gap to narrow but the native-ECE ranking to remain informative.

B.4 Baselines

All baselines use the same hidden width (64), depth (4), optimizer, and split protocol; only the propagation operator changes. MixHop (Abu-El-Haija et al. 2019), H2GCN (Zhu et al. 2020), and LSGNN (Chen et al. 2023) are compact, faithful reimplementations evaluated in the main Table 2; POLARIS attains lower ECE than all three on every dataset. Two further directions, environment-/invariance-aware propagation and spectrum-adaptive positional encodings, are orthogonal to our operator: they re-weight or re-encode the graph rather than make aggregation entropy a controllable variable, and could in principle be combined with POLARIS. We do not reimplement them here, as faithful reproduction of such methods depends on details our budget-parity harness cannot guarantee; we expect POLARIS's sparse, signed, entropy-controlled aggregation to remain complementary.

Table 3: Baseline-specific settings.

Model	Settings
MLP	one-hop feature reference (GCN with a single layer)
GAT	4 attention heads, dropout 0.5
APPNP	$K = L$ propagation steps, teleport $\alpha = 0.1$
GCNII	initial residual $\alpha = 0.1$, identity strength $\theta = 0.5$

B.5 Tuned-hyperparameter comparison

Beyond the budget-parity protocol of the main paper, we also compare the dense signed specialists under *each method's own tuned hyperparameters* (matched epochs, 3 seeds over the 10 splits) on the two most-contested WebKB graphs (Table 4). POLARIS is best on Wisconsin, statistically tied with GGCN on Texas (overlapping std), and best-calibrated on Texas; the single-seed edge the dense baselines sometimes show is largely seed noise.

Table 4: Tuned protocol: each method's own hyperparameters, matched epochs; mean±std over 3 seeds (10 splits). Accuracy and ECE (↓). Best per column **bold**, second underlined.

	Texas		Wisconsin	
Method	Acc	ECE↓	Acc	ECE↓
GGCN	**83.9±0.7**	0.135±.00	86.0±0.3	**0.106±.01**
GloGNN	81.3±1.6	0.143±.02	85.8±1.2	0.110±.01
POLARIS	83.2±1.6	**0.123±.01**	**86.3±0.8**	0.113±.01

B.6 Computational cost

Table 5 reports wall-clock cost; POLARIS's overhead over GAT/FAGCN is a constant factor, consistent with the shared per-layer complexity.

Table 5: Computational cost on Actor ($|V|$=7,600, $|E|$=30,019; depth 4, width 64, single CPU, mean over 30 steps). "ms/fwd" is inference only; all operators share the per-layer complexity $\mathcal{O}(|E|d+|V|d^2)$. **Bold**/<u>und.</u> = fastest/second among the heterophily-capable methods (GAT, FAGCN, POLARIS-U, POLARIS); GCN is shown for reference.

Method	Params (k)	ms/epoch	ms/fwd
GCN	9.5	7.2	2.4
GAT	85.1	<u>41.0</u>	<u>11.9</u>
FAGCN	77.6	**19.6**	**6.6**
POLARIS-U	78.4	43.3	17.9
POLARIS	78.8	55.4	18.7

B.7 Code and data availability

All code, trained-model configurations, and the per-split result files used to produce the tables and figures are released at https://github.com/POLARISHeterophilous/polaris.

C Additional Experiments and Analyses

The studies below extend the main-paper evaluation. Every run uses the *same* protocol, seeds, and splits as the main paper (Table 1: hidden 64, depth 4, 120 epochs, Adam lr 0.01 / weight decay 5×10^{-4}, label smoothing 0.1, dropout 0.5, 10 Geom-GCN splits, seed = split index; Cora uses its single public split). No method is tuned beyond this shared budget. All numbers are measured from real runs.

C.1 Per-node temperature and a learned signedness gate

Two natural extensions target finer-grained adaptivity than POLARIS's global τ and per-dataset signed/unsigned choice: (i) a *per-node temperature* $\tau_i = \tau_{\min} + (\tau_{\max} - \tau_{\min})\sigma(w_\tau^\top h_i)$, and (ii) a *learned per-edge signedness gate* $g_{ij} \in (0, 1)$ giving an effective coefficient $\psi_{ij}^{\text{eff}} = g_{ij}\psi_{ij} + (1 - g_{ij})$ that interpolates between the signed ($g \to 1$) and unsigned ($g \to 0$) operators per edge. Both keep $|\psi^{\text{eff}}| \leq 1$, so Theorem 1 still holds. Table 6 (with homophilous Cora as the stress test) shows the expected picture. On heterophilous graphs the plain signed operator is already best; the two adaptive variants are within 1–3 points and add no accuracy. On Cora the signed operator pays the known homophily cost (drops to 68.4 where unsigned POLARIS-U reaches 80.4); the learned gate *recovers most of this without being told the regime* (68.4 → 74.8), as does per-node τ (→ 74.0). A per-edge gate thus narrows the signed/unsigned trade-off but does not eliminate it: selecting the family per dataset still wins. Per-node τ matching global τ on heterophily is consistent with our main finding that realised entropy is driven by logit spread rather than τ (Fig. 2, main).

Table 6: Adaptive variants vs. the two base operators (accuracy %, mean±std). All build on the shared protocol. Cora is homophilous (h=0.81).

Variant	Texas	Wisconsin	Cornell	Cora
POLARIS-U (unsigned)	71.6±7.9	76.1±5.4	62.4±7.9	**80.4**
POLARIS (signed)	**80.8±4.9**	**84.7±5.1**	**75.9±6.6**	68.4
+ per-node τ	<u>79.2±9.9</u>	<u>84.3±4.8</u>	<u>74.6±5.2</u>	74.0
+ sign gate	<u>78.1±4.8</u>	<u>84.3±4.3</u>	73.2±5.3	<u>74.8</u>

C.2 Calibration: variance, binning, and temperature scaling

Table 7 reports NLL and ECE with per-split standard deviation, ECE recomputed at several bin counts, and post-hoc temperature scaling (a scalar T fitted on validation by NLL, applied to test). Three points. (i) ECE/NLL std is modest, consistent with the small-graph variance that makes WebKB differences only suggestive (significance arrives at scale on the large-graph benchmarks in the main paper). (ii) Absolute ECE grows with bin count (finer bins average less), but the dataset ordering is stable, so our fixed 15-bin choice is not a lucky operating point. (iii) The fitted T lies in $[0.72, 0.91]$ and temperature scaling does *not* materially improve ECE (ECE@15 $0.151 \to 0.149$ Texas, $0.107 \to 0.111$ Wisconsin, $0.148 \to 0.158$ Cornell): POLARIS's calibration advantage is intrinsic rather than an artifact of needing post-hoc correction.

Table 7: Calibration detail for POLARIS (signed), 10 splits. ECE@b uses b equal-width bins; T is the validation-fitted post-hoc temperature.

	NLL	ECE@10	ECE@15	ECE@20	ECE@30	T
Texas	0.643±.12	0.108	0.151	0.168	0.199	0.72
Wisconsin	0.523±.15	0.103	0.107	0.143	0.155	0.82
Cornell	0.740±.10	0.120	0.148	0.184	0.208	0.91

C.3 The label-safe bias: magnitude, effect, and a label-free proxy

The training-only discriminative bias uses labels but is disabled at inference (Prop. 5). Table 8 quantifies it. The learned magnitude λ_{disc} sits near 0.51 on all four datasets and is uncorrelated with the accuracy it buys (Pearson $r = -0.12$ between λ_{disc} and Δacc): the bias is a small, near-neutral refinement (Δacc 0.0–1.6), matching the ablation in the main paper. To test whether this benefit depends on ground-truth labels at all, we replace the labels in the bias with *pseudo-labels* (the high-confidence self-predictions of a bias-free POLARIS, a purely self-supervised structural signal). This proxy matches or exceeds the labelled bias (e.g. Wisconsin 85.9 vs. 84.7; equal on Cornell/Actor), showing the benefit does not depend on ground-truth labels and confers no unfair advantage over baselines.

Table 8: Label-safe bias on the heterophilous suite. h = edge homophily; λ_{disc} = mean learned bias strength; acc with / without the bias and with a pseudo-label proxy (accuracy %).

	h	λ_{disc}	bias on	bias off	Δ	pseudo
Texas	0.11	0.510	80.8	80.8	+0.0	80.0
Wisconsin	0.20	0.507	84.7	84.1	+0.6	85.9
Cornell	0.13	0.513	75.9	74.3	+1.6	75.9
Actor	0.22	0.519	35.4	35.3	+0.1	35.4

C.4 Signed-vs-unsigned selection by validation

The signed/unsigned choice is not a hidden hyperparameter: it is made by the same validation accuracy that selects every checkpoint. Table 9 reports how often validation prefers signed and the test accuracy of always-signed, always-unsigned, the validation-selected variant, and the per-split oracle. Validation picks signed on every heterophilous split but one (39/40 WebKB+Actor) and unsigned on homophilous Cora, and the selected accuracy equals the oracle on four of five datasets (off by 1.3 on Cornell): the knob is set reliably and at near-oracle cost.

Table 9: Validation-based selection of the signed/unsigned variant. "pick=sign" counts splits where validation prefers signed; remaining columns are test accuracy (%): always-signed, always-unsigned, validation-selected, and the per-split oracle.

Dataset	pick=sign	always-S	always-U	val-sel	oracle
Texas	10/10	80.8	71.6	80.8	81.1
Wisconsin	10/10	84.7	76.1	84.7	84.7
Cornell	9/10	75.9	62.4	74.6	75.9
Actor	10/10	35.4	33.9	35.4	35.4
Cora	0/1	68.4	80.4	80.4	80.4

C.5 What carries the same-class attention mass

A natural diagnostic for the class-mean intuition is the same-class attention mass p versus the edge-homophily chance level h. We find p sits far above h ($p-h$ from $+0.40$ to $+0.49$), but decomposing it shows this gap is *self-retention*, not selective attention to same-class neighbours. Table 10 separates the same-class mass on the augmented neighbourhood (**p_incl**) into the self-loop's own share (**self_mass**) and the same-class mass over *real* neighbours (**p_excl**). The self-loop alone holds about half the attention mass, and on real neighbours same-class attention is at or slightly below chance ($p_{\text{excl}} - h < 0$) on every dataset. The pattern is stable across depth (Actor's four layers vary by < 0.01 in each quantity) and across classes ($p_{\text{excl},c} \approx h_c$ within ± 0.002 for every Actor class). This matches the operator's design: under heterophily POLARIS leans on the node's own features (self-mixing, Prop. A2) and uses the *signed* coefficient to repel disagreeing neighbours, rather than concentrating attention on the rare same-class ones.

Table 10: Decomposition of the same-class attention mass (POLARIS signed, inference, layer-averaged over 10 splits). h: edge homophily; p_incl: same-class mass incl. self-loop; self_mass: self-loop's share; p_excl: same-class mass over real neighbours.

Dataset	h	p_incl	self_mass	p_excl	p_excl$-h$
Texas	0.108	0.602	0.585	0.026	-0.082
Wisconsin	0.196	0.617	0.530	0.146	-0.050
Cornell	0.131	0.607	0.553	0.093	-0.038
Actor	0.219	0.616	0.510	0.153	-0.066

C.6 Entropy controllability and POLARIS-E

POLARIS-E adds the aggregation-entropy regulariser, making the realised entropy a *set* quantity. Table 11 fixes distinct normalised entropy targets ρ and reports the realised entropy, accuracy, and ECE: the realised entropy tracks the target across its whole range (ρ=0.3 $\to H \approx$ 0.2, ρ=0.9 $\to H \approx$ 0.8), confirming the control is real, and lower targets modestly help accuracy/calibration. Table 12 gives the standalone POLARIS-E results (entropy regulariser on, target learned) against the unregularised POLARIS-U: accuracy is essentially unchanged on the heterophilous datasets, the realised entropy is pulled to a controlled ≈ 0.5 (versus POLARIS-U's free 0.7–1.0), and calibration improves on homophilous Cora (ECE 0.089 vs 0.117). POLARIS-E is thus an analysis/control variant: its value is the settable entropy, not an accuracy gain, exactly as positioned in the main paper. In practice we recommend POLARIS-E when a known, controlled entropy is wanted (e.g. calibration-sensitive deployments, or to diagnose the selectivity regime), with a per-node target as the better default; otherwise the free-temperature POLARIS/ POLARIS-U is preferred, as the temperature is not the lever that drives accuracy.

Table 11: Entropy controllability (POLARIS-E with fixed target ρ; realised normalised entropy, accuracy %, ECE; mean over 10 splits).

	ρ	realised H	acc	ECE
Texas	0.3	0.211	**74.1**	**0.164**
	0.5	0.448	70.8	0.175
	0.7	0.640	70.3	0.192
	0.9	0.768	<u>72.2</u>	<u>0.166</u>
Wisconsin	0.3	0.242	**78.2**	**0.134**
	0.5	0.481	76.9	0.144
	0.7	0.690	76.9	<u>0.142</u>
	0.9	0.794	<u>77.1</u>	**0.134**
Cornell	0.3	0.233	**65.4**	**0.201**
	0.5	0.469	62.2	0.247
	0.7	0.661	<u>62.4</u>	<u>0.224</u>
	0.9	0.789	60.8	0.227

C.7 Granularity of the signedness gate

Beyond the per-edge gate of Sec. C.2, Table 13 compares learning the signed$\leftrightarrow$unsigned blend at edge, node, or

Table 12: Standalone POLARIS-E (entropy regulariser on, target learned) vs. unregularised POLARIS-U, shared protocol, 10 splits (Cora: 1). Accuracy %, ECE, NLL, and realised normalised aggregation entropy H.

Metric	Variant	Texas	Wisc.	Cornell	Actor	Cora
Acc	POLARIS-U	**71.6**	76.1	62.4	**33.9**	**80.4**
	POLARIS-E	68.6	**76.7**	**62.7**	33.7	72.8
ECE	POLARIS-U	**0.175**	**0.123**	**0.223**	**0.077**	0.117
	POLARIS-E	0.180	0.139	0.235	0.139	**0.089**
NLL	POLARIS-U	0.856	0.700	1.179	1.493	0.701
	POLARIS-E	0.967	0.708	1.211	1.570	0.863
H	POLARIS-U	0.711	0.753	0.705	0.697	0.996
	POLARIS-E	0.503	0.530	0.551	0.515	0.705

single-per-layer scope. On heterophilous graphs all three sit 1–3 points below the plain signed operator (the gate adds capacity without accuracy there); on homophilous Cora every gate recovers most of the signed operator's homophily cost, with the *per-node* gate best (77.6 vs. signed 68.4, approaching unsigned 80.4). A simple per-node switch is thus the most effective of the three and a practical middle ground, though selecting the family per dataset (Table 9) remains the strongest option.

Table 13: Signedness-gate granularity (accuracy %, mean±std; ECE in parentheses), shared protocol. Cora is homophilous.

Dataset	signed	gate: edge	gate: node	gate: layer
Texas	**80.8 (.151)**	78.1 (.142)	77.3 (.151)	77.8 (.143)
Wisconsin	**84.7 (.107)**	84.3 (.127)	**84.7 (.111)**	84.1 (.119)
Cornell	**75.9 (.148)**	73.2 (.163)	73.8 (.175)	71.6 (.165)
Cora	68.4 (.092)	74.8 (.094)	**77.6 (.115)**	76.2 (.123)

C.8 Sensitivity to the temperature bounds

The bounds $[\tau_{\min}, \tau_{\max}]$ are fixed at $[0.25, 4]$ and not tuned. Table 14 varies them. Accuracy is insensitive (within about a point across $[0.5, 2]$, $[0.25, 4]$, and $[0.1, 10]$); wider bounds raise the reachable and realised entropy range (Prop. A6), but the model self-selects a similar operating point, so ECE moves little. The bounds are thus not a sensitive hyperparameter.

C.9 Post-hoc temperature scaling of the baselines

Section C.3 scaled POLARIS; Table 15 scales the baselines too (scalar T fitted on validation by NLL). Scaling lowers every method's ECE, confirming it is orthogonal to the operator. POLARIS's large *native* advantage narrows after scaling: it stays best-calibrated on Wisconsin (0.111) and is within 0.011 of the best on Texas (FAGCN 0.138 vs. POLARIS 0.149). The native-ECE ranking therefore overstates the post-scaling gap, as we anticipated; POLARIS's value is being well-calibrated *without* any post-hoc correction.

Table 14: Sensitivity to the temperature bounds (signed POLARIS; accuracy %, ECE, realised normalised entropy H; mean over 10 splits).

Dataset	$[\tau_{\min}, \tau_{\max}]$	acc	ECE	H
Texas	$[0.25, 4]$	80.8	**0.151**	0.654
	$[0.5, 2]$	80.5	0.167	0.596
	$[0.1, 10]$	**81.4**	0.161	0.711
Wisconsin	$[0.25, 4]$	84.7	**0.107**	0.702
	$[0.5, 2]$	84.5	0.120	0.634
	$[0.1, 10]$	**85.9**	0.119	0.734
Cornell	$[0.25, 4]$	**75.9**	**0.148**	0.678
	$[0.5, 2]$	75.4	0.163	0.579
	$[0.1, 10]$	75.1	0.149	0.740

Table 15: Post-hoc temperature scaling of baselines and POLARIS: fitted T and ECE before/after, mean over 10 splits.

Dataset	Method	T	ECE pre	ECE post
Texas	GCNII	0.83	0.243	0.202
	FAGCN	0.48	0.208	**0.138**
	LINKX	0.53	0.200	0.152
	POLARIS	0.72	0.151	0.149
Wisconsin	GCNII	0.88	0.153	0.160
	FAGCN	0.56	0.202	0.123
	LINKX	0.61	0.183	0.132
	POLARIS	0.82	0.107	**0.111**

C.10 Per-node entropy target

The entropy target ρ can be made per node (ρ_i from a linear map of h_i) instead of a single global scalar. Table 16 shows this *improves* POLARIS-E on every dataset (accuracy $+1.1$ to $+2.3$; lower ECE on Wisconsin and Cornell), at no extra asymptotic cost: finer entropy control is beneficial and a natural default for POLARIS-E.

Table 16: Per-node vs. global entropy target (POLARIS-E; accuracy %, ECE, realised normalised entropy H; mean over 10 splits).

Dataset	target	acc	ECE	H
Texas	global	68.6	0.180	0.503
	per-node	**70.3**	0.180	0.794
Wisconsin	global	76.7	0.139	0.530
	per-node	**79.0**	**0.116**	0.820
Cornell	global	62.7	0.235	0.551
	per-node	**63.8**	**0.184**	0.809

C.11 Validating the multi-class threshold $(1/K)$ on real data

The formal K-class analysis (App. A) predicts the averaged neighbour signal carries $\gamma_K(h) = (Kh-1)/(K-1)$ of μ_c, so it aligns with the true class only above the chance level

$h = 1/K$ and points *away* from it below. We confirm both the prediction and its consequence on the real datasets.

The averaged signal flips sign at $1/K$. For each graph we estimate class means from features (labels used only for this diagnostic, as in the p-vs-h probe), centre them, and measure for every node the projection of its uniform neighbour-average onto its own centred class mean, $\rho_i = \langle m_i - \bar{\mu},\ \mu_{y_i} - \bar{\mu}\rangle / \|\mu_{y_i} - \bar{\mu}\|^2$ (the real-data analogue of γ_K). Table 17 shows the mean ρ is *negative* on every dataset with $h < 1/K$ (the averaged signal points to the wrong class), essentially *zero* on Wisconsin where $h \approx 1/K$, and strongly *positive* where $h > 1/K$, the predicted sign in six of seven cases. The exception is Actor, whose raw features are barely class-informative (an MLP reaches only $\sim$30%), so the class-mean estimate is too noisy for the projection to be meaningful there.

Table 17: Real-data sign of the averaged-neighbour projection ρ onto the true class mean. Theory: $\rho < 0$ when $h < 1/K$, $\rho > 0$ when $h > 1/K$. Sign matches in 6/7 cases; Actor (weak features) is the exception.

Dataset	K	h	$1/K$	mean ρ
Roman-empire	18	0.05	0.056	−0.065
Texas	5	0.11	0.200	−0.284
Cornell	5	0.13	0.200	−0.084
Wisconsin	5	0.20	0.200	+0.007
Actor†	5	0.22	0.200	−0.391
Amazon-ratings	5	0.38	0.200	+0.706
Cora	7	0.81	0.143	+0.905

† weak features (MLP $\sim$30%); class-mean estimate unreliable.

POLARIS dominates exactly when $h < 1/K$. The same threshold predicts where POLARIS wins, which the main paper tabulates against its class count: POLARIS is the top method on every dataset with $h < 1/K$ (Roman-empire, Texas, Cornell, Wisconsin), ties at the boundary case Actor ($h \approx 1/K$), and yields to smoothing or its unsigned variant when $h \gg 1/K$ (Cora). Minesweeper is excluded there: its labels are a structural (mine-count) function rather than the class-mean model of Def. A1, so the feature-homophily threshold does not apply (POLARIS is nonetheless best there, on AUC).

C.12 Signed coefficient: attraction vs. repulsion

The signed coefficient ψ_{ij} should attract agreeing neighbours and repel disagreeing ones. Table 18 measures it at inference (layer 0). On the heterophilous WebKB graphs the learned ψ is *negative* (repulsive) and more negative on different-class edges than same-class ones (a positive gap of 0.14 to 0.31), so the operator repels disagreement more strongly than agreement; on homophilous Cora and weakly heterophilous Actor ψ is *positive* (attractive). The mean coefficient thus tracks the regime. Its correlation with raw feature dissimilarity is weak ($|r| < 0.14$), so the learned sign reflects label structure rather than a simple cosine rule.

Table 18: Mean layer-0 ψ on same- vs. different-class real neighbours (signed POLARIS, inference). Positive gap = stronger repulsion of different-class neighbours.

Dataset	ψ_{same}	ψ_{diff}	gap
Texas	−0.26	−0.57	+0.31
Wisconsin	−0.31	−0.59	+0.27
Cornell	−0.56	−0.70	+0.14
Actor	+0.21	+0.17	+0.04
Cora	+0.66	+0.71	−0.05

C.13 Sparse attention vs. signedness

Does sparsification or low entropy, rather than signedness, drive the gains? Table 19 replaces the softmax with *sparsemax* (which sets some weights to exactly zero) in the unsigned operator. Sparsemax does lower the realised entropy (e.g. $0.75 \rightarrow 0.53$ on Wisconsin) but leaves accuracy essentially unchanged versus unsigned softmax, and both stay far below the signed operator on every heterophilous graph. Sharper or sparser non-negative attention is therefore not the mechanism; the signed coefficient is, consistent with the ablation in the main paper.

Table 19: Attention normalisation ablation (accuracy %, ECE, realised normalised entropy H; 10 splits, Cora 1).

Dataset	Variant	acc	ECE	H
Texas	POLARIS-U (softmax)	71.6	0.175	0.711
	POLARIS-U (sparsemax)	68.1	0.171	0.558
	POLARIS (signed)	**80.8**	**0.151**	0.654
Wisconsin	POLARIS-U (softmax)	76.1	0.123	0.753
	POLARIS-U (sparsemax)	76.9	0.132	0.528
	POLARIS (signed)	**84.7**	**0.107**	0.702
Cornell	POLARIS-U (softmax)	62.4	0.223	0.705
	POLARIS-U (sparsemax)	61.6	0.223	0.540
	POLARIS (signed)	**75.9**	**0.148**	0.678
Cora	POLARIS-U (softmax)	**80.4**	0.117	0.996
	POLARIS-U (sparsemax)	79.9	0.124	0.944
	POLARIS (signed)	68.4	**0.092**	0.984

C.14 Depth at scale

The main depth study (Table 5) used the small Wisconsin graph; Table 20 repeats it on Actor (7.6k nodes). POLARIS dominates at the usable depths (2–4: $\sim 35\%$ vs. $\sim 28\%$ for GCN/GAT); beyond depth 4 all three operators decay to the degree-prior floor ($\sim 25\%$), where the residual differences are within split noise. This is exactly the scope of Theorem 1: non-expansiveness bounds explosion and keeps POLARIS from collapsing faster than averaging, yet it does not by itself delay oversmoothing indefinitely on larger graphs.

C.15 Multi-head attention

POLARIS's attention extends to multiple heads. Table 21 sweeps the head count (signed, shared protocol): two to four heads give a modest gain (up to +2.2 over one head) and

Table 20: Depth sweep on Actor (test accuracy %, mean over 10 splits). At depth ≥ 8 all operators sit at the degree-prior floor.

Depth	2	4	8	16
GCN	28.5	27.3	26.2	25.2
GAT	27.9	27.5	25.0	24.0
POLARIS	**35.3**	**35.4**	25.2	25.2

slightly raise the realised entropy (head averaging diversifies attention), without changing the regime behaviour. The operator analysed in the paper is the single-head case; multi-head is a cheap optional improvement that we use in the tuned setting.

Table 21: Multi-head POLARIS (signed): accuracy % (realised normalised entropy H), 10 splits.

Dataset	1 head	2 heads	4 heads
Texas	80.8 (0.65)	**82.7** (0.73)	80.0 (0.77)
Wisconsin	84.7 (0.70)	86.1 (0.77)	**86.3** (0.79)
Cornell	75.9 (0.68)	**78.1** (0.75)	76.5 (0.80)

C.16 Entropy as an accuracy–calibration control

Prop. A4 predicts that lowering the aggregation entropy raises the aggregate variance, so the entropy regulariser should not buy accuracy and, away from the accuracy-optimal entropy, should trade accuracy for a different operating point. We sweep the regulariser strength β (the entropy regulariser of the main paper) and report accuracy, ECE, and realised normalised entropy H; $\beta{=}0$ is the unregularised model.

On the signed model (heterophily). Table 22: increasing β monotonically lowers H (the control acts) but *costs* accuracy and does not improve ECE. With the sign already supplying selectivity, forcing low entropy only adds variance (Prop. A4); the two mechanisms are not complementary, so the deployed signed model uses $\beta{=}0$.

Table 22: β sweep on the *signed* model (accuracy %, ECE, realised H; 10 splits). $\beta{=}0$ is POLARIS. Best ECE per dataset in **bold**.

Dataset		$\beta{=}0$	0.5	1	2	5
Texas	acc	80.8	79.5	75.9	73.0	71.4
	ECE	0.151	0.165	**0.144**	0.180	0.181
	H	0.65	0.49	0.48	0.46	0.43
Wisconsin	acc	84.5	82.4	81.2	82.2	78.6
	ECE	**0.101**	0.151	0.149	0.145	0.118
	H	0.70	0.54	0.51	0.48	0.49
Cornell	acc	75.9	73.0	72.4	68.6	67.3
	ECE	**0.148**	0.161	0.179	0.193	0.183
	H	0.68	0.55	0.51	0.51	0.47

On the unsigned model (homophily). Table 23: here the trade-off is visible. On Cora, raising β markedly improves calibration (ECE $0.117 \to 0.053$) at an accuracy cost $(79.9 \to 73.1)$: the homophilous baseline already sits at high (accuracy-optimal) entropy (Rem. 2), so the regulariser only moves along the accuracy–calibration front. The effect is dataset-dependent (clear on Cora, marginal on the already well-calibrated Pubmed, unfavourable on Citeseer), consistent with the control being a lever, not a universal gain.

Table 23: β sweep on the *unsigned* model on homophilous graphs (accuracy %, ECE, realised H; public split, 3 seeds). $\beta{=}0$ is POLARIS-U. Best ECE per dataset in **bold**.

Dataset		$\beta{=}0$	0.5	1	2	5
Cora	acc	79.9	75.1	73.1	72.9	71.6
	ECE	0.117	0.092	**0.053**	0.111	0.125
	H	1.00	0.87	0.85	0.78	0.68
Citeseer	acc	69.1	67.7	66.5	66.7	65.1
	ECE	**0.069**	0.105	0.144	0.147	0.155
	H	0.98	0.89	0.98	0.97	0.94
Pubmed	acc	76.0	74.6	75.7	76.1	75.6
	ECE	0.045	**0.038**	0.079	0.071	0.041
	H	1.00	0.95	1.00	0.99	0.97